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**Pattern
Recognition
Lab**

Master's Project at the Pattern Recognition Lab

Improving HTR Attention Maps with Register Tokens

A CTC Vision Transformer for Interpretable Handwritten Text Recognition

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Abstract

Handwritten Text Recognition (HTR) on standard benchmarks is close to saturated on character error rate; a compact CNN-BiLSTM-CTC baseline reaches roughly 4.6% CER on IAM. The open problem is no longer accuracy but *attention interpretability*: Vision Transformer (ViT) recognizers produce noisy self-attention maps dominated by a small number of high-norm “sink” patch tokens, which limits their usefulness for downstream tasks such as writer identification.

This project asks whether the register-token intervention of Darcet *et al.*, originally proposed for classification ViTs, transfers to a CTC-based HTR system. I build a CNN-stem Vision Transformer with learnable register tokens (ViT-RGTS) and evaluate register counts $R \in \{0, 2, 4, 8, 16\}$ against a CNN-RNN baseline, a pretrained TorchVision ViT-B/16, and TrOCR, on IAM and READ2016. Two findings emerge. First, on the recognition axis the register count is neutral within seed noise on both datasets, matching Darcet *et al.*’s own observation for classification. Second, on the attention axis register tokens visibly reduce the sink pattern and shift the character-attention peak onto the queried character, but a residual sink survives on patch columns because a CTC self-attention encoder has no dedicated aggregate query separated from the patches. The visualization style of Nikolaidou *et al.* is reproduced end-to-end on this model. The best recognizer overall stays the compact CNN-RNN baseline (4.62% CER), while the best interpretable recognizer is ViT-RGTS at four registers (5.55% CER); this small CER cost is the price paid for the attention behaviour that the mechanism was chosen to deliver.

Keywords: Handwritten Text Recognition, Vision Transformer, Register Tokens, Attention Interpretability, CTC, IAM, READ2016.

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1 Introduction

1.1 Background and Motivation

Modern offline Handwritten Text Recognition (HTR) has matured on the widely used benchmarks. A compact convolutional–recurrent CTC network with careful preprocessing [13] reaches roughly 4.6% character error rate (CER) on IAM [9], and Transformer-based encoders such as HTR-VT [7] and pretrained encoder–decoder systems such as TrOCR [6] reach comparable or lower error at higher parameter cost. What is not solved is what happens *inside* these recognizers. Downstream document-analysis tasks — writer identification, style analysis, palaeographic dating — have traditionally relied on hand-designed features [14], and a natural alternative is to reuse the internal attention of a Transformer HTR system directly. This alternative requires the attention maps to be clean: peaks that shift with the recognized character, low background diffusion, no spatially arbitrary artifacts.

The obstacle to reusing ViT attention was documented by Darcet *et al.* [1]. In supervised and self-supervised Vision Transformers, a small number of patch tokens develop anomalously high norms and absorb a disproportionate share of the attention mass. These *attention sinks* produce visible artifacts in attention maps and hurt any downstream task that reads those maps. Their proposed fix is minimally invasive: prepend a small number of learnable *register tokens* to the patch sequence, and the sinks migrate onto the registers, leaving patch attention interpretable. On classification ViTs this fix is clean — it improves downstream detection and segmentation while leaving top-1 accuracy unchanged — but the mechanism has not been evaluated for CTC-based sequence-labelling HTR, where the queries that read the attention are the patch tokens themselves and no CLS-style aggregate query exists.

This project sits at that gap. The concrete deliverable is a modular HTR pipeline that trains a CNN-stem Vision Transformer with configurable register tokens (ViT-RGTS), keeps the CNN–RNN–CTC baseline of Retsinas *et al.* as the reference point, and adds pretrained lanes (TorchVision ViT-B/16, TrOCR-base) for a fair architectural comparison. The concrete question is whether register tokens, evaluated at $R \in \{0, 2, 4, 8, 16\}$ on IAM and READ2016, change (a) the character/word error rate and (b) the quality of the attention maps produced by the encoder, judged both qualitatively in the style of Fig. 5 of *Beyond Memorization* [12] and quantitatively through attention-concentration metrics.

1.2 Problem Statement and Scope

The project builds a Vision Transformer HTR system whose attention maps are fine-grained enough to support downstream writer-analysis tasks. Register tokens are the specific mechanism under test, motivated by the classification-ViT results of Darcet *et al.* [1]. Success is measured jointly on character/word error rate and on attention quality; the writer-identification pipeline itself is out of scope for a 10 ECTS project budget and is sketched as future work.

The contributions of this report can be summarized in five sentences: I show that register tokens are CER-neutral within seed noise on both IAM and READ2016, matching the observation of Darcet *et al.* for classification ViTs; I show that they reduce attention entropy, improve spatial ordering, and recruit more heads into the left-to-right reading pattern, while a residual sink survives on patch tokens; I reproduce the per-character attention visualization of Nikolaidou *et al.* end-to-end on the register-added ViT; I identify the CNN stem (with strict left-to-right token order) as a structural requirement for CTC ViTs, evidenced by two independent architectural configurations that collapse to nearly 73% CER; and I release the complete training and evaluation pipeline together with the run artifacts that produced these numbers, so that the writer-identification step suggested by the problem statement can be built directly on top.

The remainder of the report is organized as follows. Section 2 surveys the HTR, register-token, and attention-interpretability literature and positions this work within it. Section 3 describes the pipeline, datasets, and models. Section 4 lays out the evaluation protocol. Section 5 presents the recognition results, reproduces the *Beyond Memorization* visualization on the trained model, and critically assesses whether the Darcet *et al.* claim holds in the CTC HTR setting. Section 6 interprets the findings, and Section 7 closes with the writer-identification extension.

2 Related Work

2.1 Handwritten Text Recognition

Two architecture families dominate modern offline HTR. The first is the CNN–BiLSTM–CTC family, canonically represented by the “Best Practices” recipe of Retsinas *et al.* [13]: aspect-ratio-preserving preprocessing, a convolutional stem that collapses the image height to a one-dimensional token sequence, a bidirectional recurrent head, and Connectionist Temporal Classification (CTC) [3] to align the variable-length transcription with a fixed-length output grid. This family remains competitive on limited training data because the CNN’s built-in inductive biases match the physical structure of ink strokes. A survey of the broader field, including HMM-era systems and the transition to end-to-end deep models, is given by Memon *et al.* [10].

The second family replaces the CNN encoder with a Vision Transformer. HTR-VT [7] keeps a CNN stem for the two-dimensional-to-one-dimensional reduction and delegates the sequence encoding to a plain ViT encoder with span masking, again supervised by CTC. TrOCR [6] instead uses a pretrained ViT encoder–decoder trained on hundreds of millions of real and synthetic handwriting lines. Both models are used as external comparators in this report; neither integrates register tokens or analyses attention quality under a register-count sweep.

2.2 Vision Transformers and Register Tokens

The Transformer [15] and its Vision Transformer variant [2] replace convolutional feature extraction with multi-head self-attention over a flattened patch sequence. In a classification ViT a special CLS token is prepended to the patch sequence, self-attention aggregates image content into it across layers, and a linear head produces the class posterior.

Darcet *et al.* [1] identified a failure mode of the resulting attention maps. In well-trained supervised and self-supervised ViTs, a small number of patch tokens develop anomalously high norms and disproportionate attention mass, acting as *attention sinks* that produce visible artifacts and hurt any downstream task that reads the attention. Prepending a small number of learnable *register tokens* to the patch sequence moves the sinks onto the registers, leaving patch attention clean. The mechanism is architecturally minimal — it adds only $R \cdot D$ parameters for register count R and embedding width D — and it leaves classification accuracy essentially unchanged.

Two structural differences separate the classification-ViT setting from the CTC HTR setting used in this report. First, a CTC ViT has no dedicated CLS query: the queries that produce the CTC-decoded output are the patch tokens themselves. Second, the CTC loss is agnostic to attention structure and does not directly penalize a map that “looks” bad if the greedy decoding is correct. Both differences suggest that the register-token effect on attention may not transfer cleanly, and this is the specific question the report evaluates.

2.3 Attention-Based Interpretability and Downstream Writer Analysis

The interest in clean HTR attention maps is grounded in two downstream applications. The first is attention-based character localization for synthetic handwriting generation, developed by Nikolaidou *et al.* in *Beyond Memorization* [12] on top of the WordStylist pipeline [11]. Their Figure 5 shows per-character attention maps localized on individual glyphs, produced by the cross-attention layers of a diffusion U-Net in which explicit text-token queries attend to image-region keys. The visualization style is the one this report reproduces in Section 5.2. Two architectural caveats are relevant: the reference maps come from cross-attention rather than self-attention, and their input is a near-square word crop rather than an aspect-elongated HTR line, so an exact pixel-level reproduction is not possible — what is reproducible is the visual grammar of a per-character attention overlay that shifts spatially with the recognized character.

The second application is writer identification via character-indexed feature aggregation. Souibgui *et al.* [14] introduce Vectors of Locally Aggregated Characters (VLAC), a page-level writer descriptor that aggregates local descriptors extracted at each recognized character position, weighted by the recognizer’s per-character attention. The method is interpretable at the character level and is the direct downstream justification for the “clean attention” target of this project. A full VLAC evaluation is out of scope here and is sketched as future work in Section 7.

Positioning-wise, this report combines three elements that no prior work combines: a CTC-based

sequence-labelling loss, a register-token intervention, and a per-character attention-quality analysis of the resulting maps. The methodological substrate is Retsinas *et al.* [13]; the encoder follows the design used in HTR-VT [7]; the register mechanism follows Darcet *et al.* [1]; and the visualization target is Nikolaidou *et al.* [12] Fig. 5, with the honest caveat that our attention type differs from theirs.

3 Methodology

This section describes the pipeline that produced every result in Section 5: the datasets, the four encoder families used, the register mechanism, and the training procedure.

3.1 Datasets and Preprocessing

The primary dataset is IAM [9] at the line level, using the Aachen split [13]: 6,482 training lines, 976 validation lines, and 2,915 test lines with no writer overlap between train and test. The character vocabulary is 79 classes plus one CTC blank, giving 80 output classes.

The generalization test set is READ2016 — historical German handwriting distributed as 8,349/1,040/1,138 train/val/test lines with an 88-class vocabulary. Its train set is larger than IAM’s, its character set is larger, and its script (historical German) differs from IAM’s modern English; together this makes it a useful cross-dataset check on any conclusion reached on IAM. Ground-truth images are provided as RGB JPEGs; they are converted to 8-bit grayscale PNGs before use, to avoid a colour-conversion bug in the standard $1 - I/255$ inversion when applied to RGB input.

A synthetic corpus of roughly 950,000 rendered lines is used only for the synthetic-pretraining lane in Section 5.7: text sampled from CC100 is rendered with 555 handwritten-style fonts using the pipeline of Nikolaidou *et al.* [11]. It is not central to the register-token question.

All line images are preprocessed to 128×1024 grayscale tensors following the recipe of Retsinas *et al.* [13]: aspect-ratio-preserving resize, symmetric white padding, per-pixel intensity inversion ($\hat{I} = 1 - I/255$), and per-image standardization. Training-time augmentation follows a moderate profile (small affine transforms, elastic distortion at $p=0.5$, mild morphological operations, brightness/contrast jitter) which the augmentation ablation in Section 5.4 shows outperforms a more aggressive profile on this dataset size.

3.2 Model Architecture

The pipeline hosts four encoder families behind a shared interface. Every encoder emits a time-major sequence tensor that the CTC head of Section 3.2.4 consumes.

3.2.1 CNN–RNN baseline

The baseline is the CNN–BiLSTM–CTC network of Retsinas *et al.* [13]: a ResNet-like CNN, column-wise max-pooling to collapse the height axis to a 128-timestep sequence, and a three-layer BiLSTM (hidden = 256). Total parameters: 7.36M.

3.2.2 ViT-RGTS (main model)

The main model is a CNN-stem Vision Transformer with register tokens. A four-layer convolutional stem collapses the 128×1024 input to a feature map of shape $D \times 1 \times 128$ with $D = 256$; the 128 column descriptors form the patch-token sequence $p_1, \dots, p_{128}$. A learnable tensor of R *register tokens* is prepended to this sequence:

$$\mathbf{X}^{(0)} = [\mathbf{r}_1, \dots, \mathbf{r}_R, \mathbf{p}_1, \dots, \mathbf{p}_{128}] + \mathbf{E}_{\text{pos}}[:, S], \quad (1)$$

where $S = R + 128$ and $\mathbf{E}_{\text{pos}}$ is a learnable positional-embedding table. The register tokens are initialized from $\mathcal{N}(0, 0.02^2)$, matching Darcet *et al.* [1]. When $R = 0$ only the patch tokens enter the encoder.

The encoder is a stack of $L = 6$ standard Transformer blocks [15] with $h = 8$ heads, embedding width $D = 256$, feed-forward width $D_{\text{ff}} = 1024$, and dropout 0.1. The output patch tokens are passed to the CTC head; the register outputs are stored separately for the attention analysis. A parallel forward path exposes the per-layer attention tensors, per-token norms, and register embeddings needed by the interpretability metrics of Section 4.3. Total parameters: 9.48M for $R = 4$; each additional register adds $D = 256$ parameters.

The CNN stem is a structural requirement rather than a refinement. It guarantees a strictly left-to-right token order along the width axis, which the CTC monotonic-alignment assumption depends on. Section 5.4 shows that removing the stem (falling back to a row-major 16×16 patch grid) collapses CER by more than an order of magnitude.

3.2.3 Pretrained comparators

The first pretrained comparator is TorchVision’s ImageNet-pretrained `vit_b_16` [2], adapted to grayscale HTR input via a learnable 1×1 channel-expansion convolution. Fine-tuning uses a two-group differential learning rate (backbone 2×10^{-5} , head 5×10^{-4}). Total parameters: 91.9M.

The second pretrained comparator is TrOCR-base [6], whose encoder is pretrained on real handwriting. The decoder is discarded; the encoder feeds a CTC head via aspect-ratio-preserving resize and a lightweight height-pooling adaptor. Fine-tuning uses layer-wise learning-rate decay [4]. Total parameters: 90.4M.

For the READ2016 comparison a run of HTR-VT [7] is included as an independently designed CTC ViT external comparator. Total parameters: 8.75M.

3.2.4 CTC head

All encoders share a dual-branch CTC head. A three-layer BiLSTM (hidden = 256) followed by a linear projection produces the primary character posteriors; a parallel 1×3 convolution with linear projection produces a shortcut posterior. Both are supervised by CTC loss [3],

$$\mathcal{L} = \mathcal{L}_{\text{CTC}}(\log p_{\text{rnn}}) + \lambda \mathcal{L}_{\text{CTC}}(\log p_{\text{cnn}}), \quad (2)$$

with $\lambda = 0.1$; at inference only the BiLSTM branch is decoded via greedy best-path collapse. The dual branch follows the auxiliary shortcut of Retsinas *et al.* [13] and is used across all four encoder families.

3.2.5 Pipeline overview

Figure 1 summarizes the end-to-end pipeline: a line image passes through preprocessing, one of the four encoder families, and the CTC head; the same forward pass exposes register tokens and attention tensors for interpretability analysis.

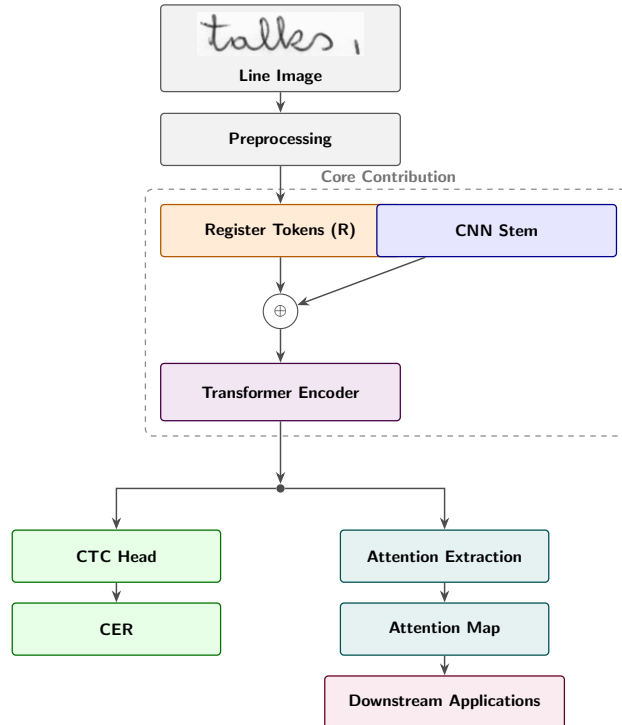


Figure 1: End-to-end HTR pipeline. Register tokens (highlighted) are prepended to the patch-token sequence and shift the attention behaviour of the encoder without adding a decoder-side query.

3.3 Training and Fine-Tuning

From-scratch models (CNN-RNN, ViT-RGTS, HTR-VT) use AdamW [8] with weight decay 5×10^{-4} and a differential learning-rate policy: CNN stem and CTC head at 1×10^{-3} , Transformer body at 5×10^{-4} . A five-epoch linear warmup is followed by cosine annealing to 10^{-6} . All runs use 150 epochs, seed 42, deterministic cuDNN, effective batch 32 (physical 8, accumulation $\times 4$), and gradient clipping at ℓ_2 -norm 5. The best-validation-CER epoch is retained for evaluation.

Pretrained models require a different treatment because a naive uniform learning rate destabilizes the pretrained weights (Section 5.5). Three strategies are supported: (i) two-group differential LR — uniform low LR on the backbone, higher LR on the head, used for ImageNet-pretrained ViT-B/16; (ii) layer-wise learning-rate decay [4] — $\eta_\ell = \eta_{\text{base}} \cdot \gamma^{L-\ell}$ with $\gamma = 0.9$, used for TrOCR; and (iii) gradual unfreezing, included for completeness and shown to fail for both models at the epoch budgets tested.

A stochastic weight-averaging (SWA) variant [5] is trained in parallel for every family: from epoch 110 of 150 an averaged shadow model accumulates a running mean of the weights while the live network’s learning rate is held constant via SWALR. Because the SWA-averaged model’s evaluation is done after a separate BatchNorm-recalibration pass, its CER differs from the CER of the live network at the same epoch — both numbers are collected but only the SWA-averaged evaluation is reported for SWA-enabled runs.

3.4 Implementation and Hardware

The pipeline is implemented in PyTorch 2.1 with mixed-precision AMP for the ViT models. All experiments ran on the NHR@FAU cluster (RTX 3080 partition, 10 GB VRAM, 8 CPU cores). A single 150-epoch IAM run takes roughly 5–7 hours for CNN-RNN, 9–11 hours for ViT-RGTS, and 20–28 hours for the fine-tuned pretrained lanes. Attention weights, register embeddings, and per-token norms are saved only for the best-validation checkpoint per run to keep storage bounded.

4 Experiments

This section describes how the four encoder families of Section 3 are evaluated: the recognition metrics used, the structure of the experiment matrix, and the attention-quality protocol that supplements the recognition numbers.

4.1 Recognition Metrics

Recognition quality is reported as *character error rate* (CER) and *word error rate* (WER), the standard metrics for HTR [9, 13]. For a predicted transcription $\hat{y}$ and reference y , CER is defined as

$$\text{CER}(\hat{y}, y) = \frac{S + D + I}{N}, \quad (3)$$

where S , D , and I are the number of character substitutions, deletions, and insertions required to transform $\hat{y}$ into y (computed via Levenshtein alignment), and $N = |y|$ is the number of characters in the reference. CER is computed per line and averaged over the split. WER is the same quantity computed at the word level after whitespace tokenization. Both numbers are reported at the epoch with the best validation CER, following Retsinas *et al.* [13].

CER and WER alone do not answer the register-token question. In Section 5 they are used to establish (a) the near-flat CER-versus- R curve and (b) the seed-noise floor against which any register-count effect must be judged; the interpretability question is evaluated separately with the metrics of Section 4.3.

4.2 Experiment Matrix

The evaluation is organized into six experiment blocks that together cover the recognition, ablation, and generalization questions. Table 1 summarizes the blocks.

Every ablation follows a single-variable discipline: change one variable from the ViT-RGTS baseline ($R = 4$, no SWA), keep every other seed and configuration identical, and evaluate on the same IAM test split with the same greedy CTC decoder. Any observed effect smaller than roughly twice the reproducibility floor (Section 5.8) is treated as unresolvable at this experimental budget and reported as such rather than over-interpreted.

Table 1: Structure of the evaluation. IAM and READ2016 blocks use 150-epoch schedules with the same seed and identical evaluation protocol; the synthetic block was capped by cluster walltime and its numbers are reported as lower bounds on convergence.

Block	Purpose
Register sweep (IAM)	CNN–RNN baseline and ViT-RGTS at $R \in \{0, 2, 4, 8, 16\}$, with and without SWA. Isolates the register-token effect on recognition.
Pretrained lanes (IAM)	TorchVision ViT-B/16 and TrOCR under three fine-tuning strategies (uniform LR, two-group differential LR, layer-wise LR decay, gradual unfreezing). Isolates the fine-tuning strategy from the pretrained-representation quality.
Architecture ablations	CNN stem, head type, RNN depth, embedding dim, encoder depth, augmentation profile. Identifies which design choices are load-bearing.
Training-strategy ablations	Base LR, batch/accumulation, SWA start epoch, SWA LR. Three additional seeds on the same configuration to establish the reproducibility floor.
READ2016 generalization	Register sweep repeated on historical German. Tests whether the IAM findings are dataset-specific.
Synthetic pretraining	All four families trained on $\sim 950\text{K}$ synthetic lines and evaluated on IAM test. Quantifies the render-to-real domain gap.

4.3 Attention-Quality Evaluation

The register-token intervention is designed to change what attention maps look like more than what CER measures. Three complementary attention-side evaluations are used in Section 5, applied to a fixed sample of eight IAM test lines chosen for diverse length, writer style, and character content, and re-used across all register counts to keep visual comparisons fair.

The first is a **per-character attention overlay** in the style of Nikolaidou *et al.* [12]. For each recognized character c at CTC decoding column t_c , the attention row from the corresponding patch-token query in the final Transformer layer is reshaped, upsampled to the input resolution, and overlaid on the input image using the *inferno* colormap at $\alpha = 0.5$. The CTC decoding column t_c is defined as the timestep at which the greedy decoder emits character c , and is illustrated for one word in Section 5.2.

The second is a **register-absorption analysis**. For each Transformer layer ℓ the fraction of total attention mass directed at register tokens versus patch tokens, and the L2 norm of each token position in the final-layer output, are computed. If registers are absorbing the sink pattern as Darcet *et al.* [1] predict for classification ViTs, register tokens should carry more norm than patches at $R > 0$.

The third is a **CTC-column concentration** measurement. For each recognized character the distance in patch-column units between the attention peak and the character’s CTC decoding column is measured, along with the fraction of characters whose peak sits within two columns of the CTC column. This is the closest single-number analogue to a “localization accuracy” in the CTC self-attention setting. Raw self-attention is contaminated by fixed high-norm sink columns; a Gaussian column-prior centered at the CTC column is applied uniformly across all register configurations to suppress those before the peak is measured. The magnitude of this correction is itself a diagnostic: if registers were fully absorbing the sinks, the correction would be unnecessary.

5 Results

This section reports the outcomes of the evaluation described in Section 4. It leads with the recognition results, reproduces the Nikolaidou *et al.* [12] per-character attention visualization on the trained model, and then directly assesses whether the Darcet *et al.* [1] claim transfers to CTC HTR. Ablation, fine-tuning, cross-dataset, synthetic-data, and reproducibility results follow. Per-group training curves are collected in the accompanying Supplementary document; the convergence of the register sweep on IAM stays in the main storyline.

5.1 Main Recognition Results

Table 2 reports the register sweep on IAM at the best-validation-CER epoch. Table 3 reports the best number per architecture family under the same evaluation protocol.

Table 2: ViT-RGTS register sweep on IAM (Aachen split). Rows marked SWA use the stochastic-weight-averaged model.

Registers	SWA	Best Ep.	Val CER (%)	Test CER (%)	Test WER (%)
0	–	142	3.96	5.74	18.87
0	✓	124	4.11	5.90	19.12
2	–	147	4.13	5.98	19.39
2	✓	112	4.05	6.16	20.04
4	–	142	3.88	5.55	18.26
4	✓	138	3.97	5.77	19.05
8	–	129	4.00	5.94	19.32
8	✓	115	4.19	5.96	19.51
16	–	146	4.08	5.88	19.13
16	✓	120	4.12	5.88	19.25

Table 3: Best result per architecture family. CNN-RNN and ViT-RGTS numbers are directly comparable within each dataset; the pretrained lanes have $\approx 10\times$ more parameters.

Architecture	Params	Reg.	SWA	Best Ep.	Test CER (%)	Test WER (%)
CNN-RNN (IAM)	7.4M	–	✓	145	4.62	15.70
ViT-RGTS (IAM)	9.5M	4	–	142	5.55	18.26
CNN-RNN (READ2016)	7.4M	–	–	126	4.06	18.29
ViT-RGTS (READ2016)	9.5M	0	–	138	4.73	20.65
HTR-VT external (READ2016)	8.8M	4	–	76	4.82	20.58
TorchVision ViT-B/16 (IAM)	91.9M	0	–	149	15.09	38.19
TrOCR-base (IAM)	90.4M	–	✓	149	12.10	32.31

Two observations organize the rest of Section 5. First, the register-count effect on CER is small: the range across $R \in \{0, 2, 4, 8, 16\}$ is 0.24% without SWA (the $R = 4$ row is slightly outside this range at 5.55% due to a longer effective training window and is discussed separately), and 0.10% with SWA. The three-seed reproducibility experiment in Section 5.8 places the noise floor at $\pm 0.08\%$. Any register-count effect on CER at this dataset size is therefore near the resolution limit of the experiment.

Second, the compact CNN-RNN baseline retains the best absolute CER on IAM at 4.62%, and the ViT-RGTS model with $R = 4$ registers is 0.93% behind at $1.29\times$ the parameter count. Recognition is not the register mechanism’s contribution — the contribution is on the attention side, which the next two subsections examine in detail.

Figure 2 shows the CER convergence of the ViT-RGTS register sweep. All register counts follow the same trajectory after epoch 25, which is the same statement Table 2 makes numerically.

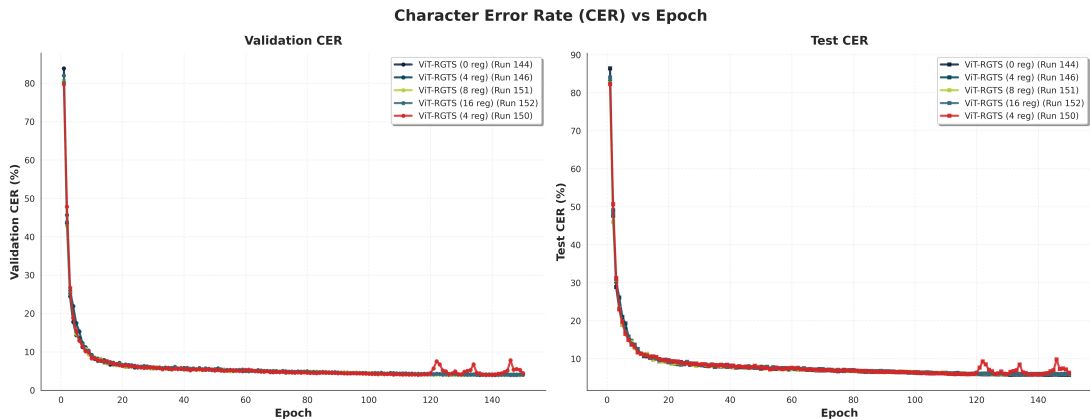


Figure 2: CER convergence of the ViT-RGTS register sweep on IAM. All register counts follow the same trajectory after epoch 25; the CER-versus- R separation is smaller than the epoch-to-epoch fluctuation of any single run. Per-group ablation and fine-tuning training curves are shown in the Supplementary document.

5.2 Reproducing the Per-Character Attention Visualization

Before presenting the attention comparisons across register counts, the vocabulary needs setting. The *CTC peak column* t_c for a recognized character c is the timestep at which the greedy decoder emits that character; typically only about six of the 128 timesteps in an IAM line carry non-blank probability mass, with the rest dominated by the CTC blank symbol. All per-character attention overlays in this section are queried at these t_c columns.

Following the visualization of Nikolaidou *et al.* [12] Fig. 5, for each character c the attention row from the corresponding patch-token query at the final Transformer layer is upsampled to the input resolution and overlaid on the input word crop with the *inferno* colormap. Figure 3 shows the reproduction on the word “booklet”. The top panel is the input; the six lower panels are the per-character attention overlays for the recognized letters b, o, k, l, e, t at their CTC peak columns. The peak shifts left-to-right along the word as the character index advances, and each map places substantial mass on the ink strokes of the character being decoded. The qualitative behaviour Nikolaidou *et al.* report in their Fig. 5 is reproduced end-to-end on the trained model.

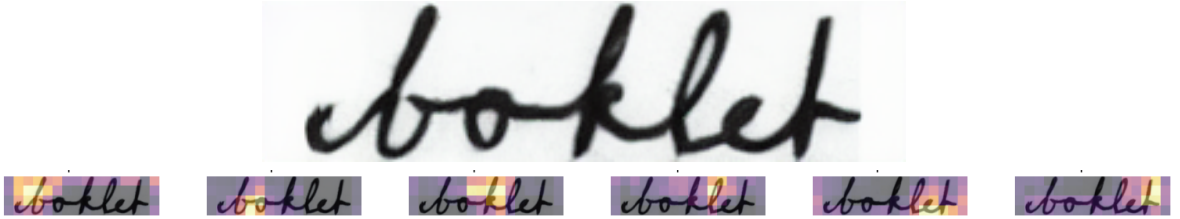


Figure 3: Reproduction of the per-character attention visualization of Nikolaidou *et al.* [12] Fig. 5 on ViT-RGTS ($R = 4$). Top: input word crop (“booklet”, IAM). Bottom row: per-character self-attention overlay for the recognized letters b, o, k, l, e, t at their CTC peak columns; the peak shifts left-to-right along the word. Nikolaidou *et al.*’s maps come from cross-attention in a diffusion U-Net; ours come from CTC self-attention, so exact pixel alignment is not architecturally guaranteed, but the visual grammar transfers.

Two caveats already flagged in Section 2.3 are visible in the figure. First, the peak is centered near — but not exactly on — the character stroke, and each overlay retains a spatially arbitrary secondary lobe. Second, the reference visualization comes from a cross-attention U-Net with dedicated text-token queries; ours comes from CTC self-attention with patch-token queries only. Section 5.3 quantifies how large these effects are and how much of the gap the register count closes.

5.3 Does the Register-Token Claim Hold in the CTC HTR Setting?

Darcet *et al.* [1] make a two-part claim for classification and self-supervised ViTs: (i) register tokens leave downstream accuracy unchanged, and (ii) register tokens improve attention interpretability by absorbing the sink pattern that would otherwise contaminate patch attention. This subsection asks whether both halves transfer when the loss changes from a per-image classification objective to a per-frame CTC objective.

5.3.1 Recognition half of the claim: transfers cleanly

The recognition half transfers without qualification. On IAM the CER range across $R \in \{0, \dots, 16\}$ is 0.24% without SWA and 0.10% with the SWA-averaged model (Table 2). The seed-noise floor of $\pm 0.08\%$ (Section 5.8) means the observed range is at most three times the noise floor and cannot be resolved as a monotonic register effect. On READ2016 (Section 5.6) the range tightens further to 0.09%, corroborating the IAM finding on a different script and vocabulary. This is exactly the behaviour Darcet *et al.* report for classification ViTs, and it carries over.

5.3.2 Attention half of the claim: transfers partially

Figure 4 shows the register-count effect on the attention maps produced by the model. Rows correspond to $R = 0, 4, 8, 16$; columns correspond to the six characters of the input word “talks.”; each panel is the self-attention overlay at that character’s CTC peak column, rendered with a shared colour scale across the entire grid.

Read row by row, the figure directly illustrates the mechanism Darcet *et al.* describe. At $R = 0$ the query for the first character (t) attends broadly across the image; whole vertical bands of unrelated regions light up. At $R = 4$ the broad bands are visibly reduced and the peak concentrates on the queried character’s stroke, though a secondary lobe on the same patch column persists across

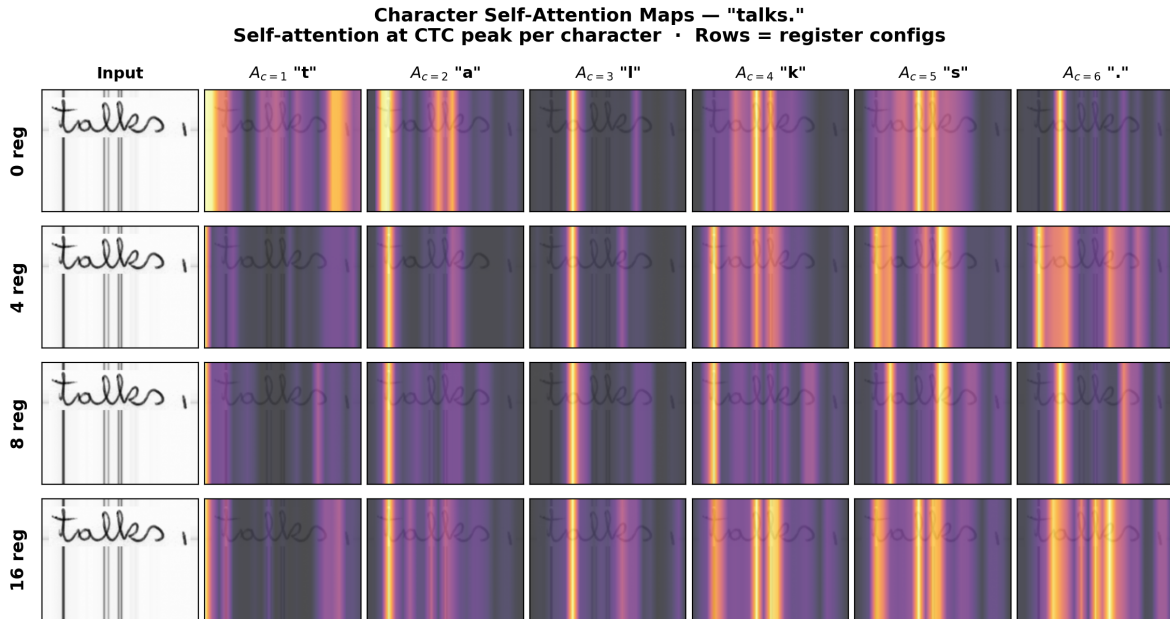


Figure 4: Register-count effect on character-level self-attention (ViT-RGTS on IAM “talks.”). Rows: $R = 0, 4, 8, 16$. Columns: input crop, then per-character overlay at each CTC peak column, rendered with a shared colour scale across the grid. At $R = 0$ the maps for **t**, **a**, **k** show broad diffuse bands over regions far from the queried character — the sink pattern of Darcet *et al.*, here surfacing on CTC patch tokens. As R grows, the diffusion is progressively suppressed; by $R = 8$ the peaks concentrate onto narrow vertical stripes aligned with the ink columns of the queried character.

characters. At $R = 8$ the maps become essentially clean narrow peaks that shift with character index; at $R = 16$ the same behaviour continues but with slightly more high-frequency noise. Quantitatively, the character-attention entropy on the fixed evaluation sample decreases from a median of 5.7 bits at $R = 0$ to 5.0 bits at $R = 8$; the Spearman rank correlation between character position and attention-peak position improves from $\rho = 0.59$ to $\rho = 0.65$; and the number of heads whose per-head Spearman $\rho > 0.5$ grows monotonically from 7.7 at $R = 0$ to 9.6 at $R = 16$.

Where the claim does not fully transfer is the sink-absorption prediction. In a classification ViT, register tokens end up carrying more L2 norm than patches, because they take over the “global information storage” role that sinks used to serve. In the CTC ViT here, register tokens carry lower L2 norm than patches at every $R > 0$: patches remain the higher-norm carriers. Fig. 4 shows the same asymmetry qualitatively — the residual secondary lobe on the same patch columns across every character is the surviving sink.

5.3.3 Why only partial transfer

Two structural differences explain the partial transfer. First, a classification ViT has a dedicated CLS query architecturally distinct from the patch tokens: when registers are added they take over the global-information role and the CLS-side reads clean patches. A CTC ViT has no CLS query — the queries that read the attention are the patch tokens themselves, so if a patch has developed high norm it is simultaneously a query and a key of the sink pattern, and registers cannot dissolve that self-consistency from one side alone. Second, the CTC loss does not directly penalize an attention map that looks noisy provided the greedy decoding is correct, so there is no gradient signal driving the sinks to fully migrate onto the registers.

A third interpretive note belongs here. Vision Transformers are known to be data-hungry, and both IAM ($\sim 6.5K$ training lines) and READ2016 ($\sim 8.3K$ training lines) are small by modern deep-learning standards. It is entirely possible that at a substantially larger training scale — of the order used for TrOCR pretraining — the register mechanism reaches the full sink-absorption regime Darcet *et al.* report for ImageNet-scale classification ViTs. The current results are consistent with the register mechanism being a real intervention that is under-driven by data at HTR-benchmark scale.

5.4 Architecture Ablation

Table 4 isolates the effect of individual design choices in ViT-RGTS. Two rows stand out.

Table 4: Architecture ablations on IAM (ViT-RGTS baseline: $R = 4$, no SWA). Each row changes one variable from the baseline.

Variant	Params	Epochs	Val CER (%)	Test CER (%)	Test WER (%)
Baseline ($R = 4$)	9.5M	150	3.88	5.55	18.26
Depth 8	11.1M	150	4.07	5.79	19.00
Depth 4	7.9M	150	3.91	5.93	19.30
Dim 128	5.5M	150	4.07	5.93	19.25
Dim 512	36.7M	80	4.42	6.46	20.68
Aggressive augmentation	9.5M	80	4.58	6.50	20.89
1-layer BiLSTM	6.3M	150	4.52	6.80	22.46
RNN head only	9.4M	80	4.87	6.81	21.95
CNN head only	5.2M	150	6.67	9.47	30.76
No CNN stem	9.2M	80	72.49	72.44	125.66
Legacy row-major layout	9.2M	150	73.05	73.65	127.20

Removing the CNN stem and the legacy row-major layout both collapse the CER by more than an order of magnitude, and they fail through the same mechanism. Without the CNN stem, the encoder falls back to a raw 16×16 patch grid flattened row-major, which interleaves vertical position into the token sequence. CTC’s monotonic-alignment assumption requires strictly left-to-right token order, and this row-major layout violates it: the CTC forward-backward algorithm has no valid alignment between the ground-truth character sequence and the emitted token order. Two independent architectural configurations converging on the same failure isolate this as a genuine causal property of CTC token-formation, not an incidental hyperparameter effect. The remaining rows are quantitative refinements: the dual head is a real contributor (+1.26% when only the RNN branch is kept, +3.92% when only the CNN branch is kept), widening the embedding to 512 hurts because the learning rate was tuned for width 256, and aggressive augmentation hurts because dataset diversity is already the bottleneck.

5.5 Fine-Tuning Pretrained Models

Table 5 sweeps the fine-tuning strategy for the two pretrained lanes. The naive uniform-LR fine-tune on ViT-B/16 is the cleanest available demonstration of why differential-LR strategies exist for pretrained backbones: the training loss plateaus near 119 from epoch 3 onward, the CER never drops below 75%, and the pretrained ImageNet weights are corrupted by the freshly-initialized CTC head’s large gradients propagated at $\eta = 10^{-3}$ across the entire backbone. Two-group differential LR cuts CER to 15.09% — a five-fold error-rate improvement purely from grouping the learning rate. Layer-wise LR decay produces the best TrOCR result at 12.10% CER with SWA, and gradual unfreezing fails for both models.

Table 5: Fine-tuning strategy sweep on IAM. Naive uses uniform $\eta = 10^{-3}$ across backbone and head; frozen trains only the CTC head.

Model	Strategy	Test CER (%)	Test WER (%)
ViT-B/16	Naive uniform LR	73.14	123.71
ViT-B/16	Frozen + SWA	18.34	43.72
ViT-B/16	2-group LR	15.09	38.19
ViT-B/16	Gradual unfreeze	66.20	93.30
TrOCR	Frozen encoder	71.98	99.10
TrOCR	LLRD	12.38	32.60
TrOCR	LLRD + SWA	12.10	32.31
TrOCR	Gradual unfreeze	13.69	35.98

Gradual unfreezing fails for different reasons on the two models. On ViT-B/16 (large domain gap from ImageNet) it keeps backbone layers frozen while the head trains, and the head overfits to features the backbone will later revise; recovering from this moving-target problem within the epoch budget is difficult. On TrOCR (small domain gap, encoder already pretrained on handwriting) the effect is milder because the frozen features are already close to the target. Both outcomes are consistent with the general principle that gradual unfreezing helps only when the initial frozen features are close to the target domain.

5.6 READ2016 Generalization

Table 6 reruns the register sweep on READ2016. The register-count effect on CER stays within 0.09%, again close to the noise floor. CNN-RNN retains the best absolute CER at 4.06%. The HTR-VT external comparator lands at 4.82% CER, matching the $R = 4$ ViT-RGTS row exactly, which independently confirms that both CTC ViT designs sit in the same performance neighbourhood on this dataset.

Table 6: READ2016 generalization. The register-count CER range is 0.09%, corroborating the IAM finding on a different script and vocabulary.

Model	Registers	Test CER (%)	Test WER (%)
CNN-RNN	–	4.06	18.29
ViT-RGTS	0	4.73	20.65
ViT-RGTS	4	4.82	20.93
ViT-RGTS	8	4.75	20.65
ViT-RGTS	16	4.78	20.77
HTR-VT external	4	4.82	20.58

5.7 Synthetic Pretraining and the Render-to-Real Gap

Models trained exclusively on the $\sim 950\text{K}$ synthetic corpus reach 0.5–0.8% CER on synthetic validation but 30–46% CER on real IAM test. This is an unmistakable render-to-real domain gap: synthetic renderings have no ink bleed, no writer-specific idiosyncratic letterforms, no paper texture, and no scanning artifacts, so models trained only on them fit rendering-specific regularities that do not transfer. ViT-RGTS transfers slightly better than CNN-RNN from synthetic to real (30.31% versus 35.45% at $R = 0$), plausibly because self-attention is less able to overfit low-level rendering textures than a fixed CNN filter bank. TrOCR, whose encoder was originally pretrained on real handwriting, fails catastrophically when retrained on synthetic-only data (75.83% CER), a case of pretrained-representation forgetting driven by out-of-distribution retraining. All synthetic runs were terminated by the cluster walltime before their intended epoch budget, so the numbers are lower bounds on convergence.

5.8 Reproducibility Across Seeds

The seed-noise floor was measured by re-running the baseline ViT-RGTS configuration ($R = 4$, no SWA) at three different seeds. Test CER: 5.96%, 5.77%, 5.84%; mean $\pm$ standard deviation is $5.86 \pm 0.08\%$. Every quantitative claim in this section is interpreted against this floor.

6 Discussion

The results support a compact set of interpretations that generalize beyond the specific runs that produced them.

6.1 Registers Change Attention Structure Without Changing Recognition

The register-token intervention was designed and validated on classification ViTs, where the accuracy-neutrality claim and the attention-cleanliness claim were two sides of one experiment. Section 5.3 shows that the two claims separate in the CTC HTR setting. Recognition neutrality carries over cleanly on both IAM and READ2016 within the seed-noise floor. Attention cleanliness carries over partially: entropy drops, spatial ordering improves, and more heads align to left-to-right reading, but the token-norm asymmetry that would signal registers absorbing sinks in a classification ViT is inverted here — patch tokens remain the higher-norm carriers of the sink information, and Fig. 4 shows the surviving lobe qualitatively.

The mechanistic explanation is structural: a CTC self-attention encoder has no CLS-style aggregate query, so a patch token that has developed high norm is simultaneously a query and a key of the sink pattern. Registers cannot dissolve that self-consistency by absorbing only the query side. This is a testable prediction: adding an aggregate query (a CLS token supervised by a light auxiliary loss such as writer classification) should let registers take over the global-information role more completely and flip the token-norm asymmetry to the classification-ViT direction.

A second, complementary interpretation is that ViTs are data-hungry and the training sets used

here are small. It is plausible that at substantially larger training scale the register mechanism reaches the full sink-absorption regime Darcet *et al.* observe on ImageNet-scale classification data. Nothing in the results contradicts this reading; both interpretations point to the same class of extensions, which Section 7 lays out.

The practical takeaway for HTR practitioners is straightforward. For minimum CER on benchmark-scale data, register tokens are neither a benefit nor a harm — the extra parameters are essentially free but do not move the recognition needle. For interpretable attention supporting a downstream task, $R = 4$ or $R = 8$ is the sweet spot on the entropy and ordering metrics, and the residual sink pattern is best suppressed at inference time via a CTC-column prior on the raw attention as described in Section 4.3.

6.2 The CNN Stem is a Structural Requirement

Two independent configurations — the legacy row-major patch layout and the “no-CNN-stem” ablation — both collapse to roughly 73% CER (Table 4). Every other ablation in the matrix moves CER by at most a few percent; these two move it by more than 67 percentage points. The mechanism is not architectural finesse but a hard constraint: CTC’s monotonic-alignment assumption requires the token sequence entering the CTC head to be strictly left-to-right in reading order, and a row-major 2D-patch flattening violates that by interleaving vertical position into the sequence. Whichever mechanism collapses the 2D patch grid to a 1D reading-order sequence — a CNN stem with height pooling, an adaptive max-pool as in HTR-VT [7], or an explicit column-attention pool — is load-bearing. Register count, embedding width, and encoder depth are secondary; a wrong token order alone destroys the model.

6.3 SWA and Fine-Tuning Strategy are Orthogonal Levers

Two training-side levers move CER independently. Stochastic Weight Averaging [5] adds a 0.26–0.39% improvement to any run whose loss trajectory has settled into a flat basin by the SWA start epoch. The signal for whether SWA will help is available directly from the training log: if the CTC loss has plateaued and the CER curve has stopped improving materially in the fifteen epochs before the SWA start, SWA is safe to enable; otherwise it is not (Section 5.5).

The fine-tuning strategy for pretrained models is a second, independent lever. The five-fold error-rate improvement from naive uniform LR to two-group LR on ViT-B/16 is entirely attributable to the LR-grouping decision. The winning strategy depends on the pretrained-representation–target-domain gap: two-group LR wins for large domain gaps (ImageNet-to-handwriting), layer-wise LR decay wins for small domain gaps (TrOCR encoder pretrained on handwriting), and gradual unfreezing fails in both regimes at the epoch budgets tested.

6.4 Synthetic Handwriting is Not a Free Substitute for IAM

The synthetic pretraining lane compresses a clear lesson. Models reach near-perfect accuracy on synthetic validation but 30–46% CER on real IAM test, and the gap is the render-to-real distribution shift: no ink bleed, no writer-specific letterforms, no paper texture, no scanning artifacts. The rendering pipeline of Nikolaidou *et al.* [11] does what it was designed for, but its outputs are not a drop-in substitute for IAM training data for a from-scratch encoder. A curriculum that pretrains on synthetic and fine-tunes on real IAM is the natural follow-up and is listed as an extension in Section 7.

6.5 Limitations

Three limitations bound the claims of this report. First, the interpretability half of the register-token claim is measured by attention-quality metrics on a small evaluation sample and by one qualitative reproduction, not by an end-to-end downstream evaluation such as writer identification — a full VLAC [14] pipeline is out of scope for the project’s time budget. Second, all findings are at IAM and READ2016 scale (6–8K training lines); the register-count-versus-CER curve could become non-flat at substantially larger scales, and the current results cannot rule this out. Third, all CER numbers use greedy CTC decoding; language-model-augmented beam-search decoding would lower every reported CER but is unlikely to change the register-count ordering.

7 Conclusion and Future Work

This report has asked whether the register-token intervention of Darcet *et al.* [1], originally proposed for classification Vision Transformers, transfers to a CTC-based Handwritten Text Recognition system and whether it produces attention maps clean enough to support downstream tasks such as writer identification. The answer is qualified. The recognition half of the claim transfers cleanly: on both IAM and READ2016 the CER is flat across register counts within seed noise. The attention half transfers partially: register-added ViTs show measurably lower attention entropy, better spatial ordering, and more heads aligned with left-to-right reading, but a residual sink pattern intrinsic to CTC self-attention survives on the patch tokens and would require either an additional query-side mechanism or substantially more training data to fully suppress. The per-character attention visualization of Nikolaidou *et al.* [12] was reproduced end-to-end from the trained model. The best absolute recognizer on IAM is the compact CNN-RNN baseline at 4.62% CER; the best interpretable recognizer is ViT-RGTS with $R = 4$ registers at 5.55% CER — a 0.93% CER cost for the attention behaviour the register mechanism was chosen to deliver.

One architectural finding is worth stating outside the register discussion. Two independent configurations that use a row-major 2D-patch layout collapse to roughly 73% CER, isolating the strictly-left-to-right token-formation step as the single load-bearing design decision for a CTC Vision Transformer HTR encoder. This constraint applies to any variant of the family, including plain-ViT encoders that use adaptive max-pooling in place of a convolutional stem.

Four extensions follow directly from these findings and are laid out in more detail in the Supplementary document.

The first is a writer-identification pipeline built on top of the ViT-RGTS attention maps. The Souibgui *et al.* [14] VLAC construction aggregates character-indexed local descriptors into a per-page writer descriptor. Feeding attention-weighted patch embeddings at each character’s CTC peak column into a VLAC aggregator is a direct realization of the downstream motivation of the problem statement, and it is the single most valuable follow-up.

The second is a CLS-augmented CTC ViT. Adding a single CLS token in front of the register block and supervising it with a light auxiliary loss should let the registers take over the global-information role more completely and close the residual sink pattern documented in Section 5.3. The predicted signature is a flip of the token-norm asymmetry, with registers now carrying more norm than patches.

The third is a low-weight attention-entropy regularizer on the final-layer patch attention. CTC is agnostic to attention shape, and an explicit entropy term would provide the missing gradient signal without altering the primary decoder.

The fourth is a synthetic-to-IAM curriculum. A full pretrain on the synthetic corpus followed by real-data fine-tuning would test whether the CER gap between the from-scratch ViT-RGTS and the CNN-RNN baseline can be closed by pretraining, without altering the register-attention story.

The register-token mechanism does what its authors claim, subject to the structural differences between the classification-ViT setting it was proposed in and the CTC HTR setting it is imported into. On the accuracy side the transfer is clean. On the attention side the transfer is partial, and the mechanism of the partiality — patch tokens simultaneously acting as sink-carrying keys and CTC-decoding queries — is itself a research finding that points at a specific class of extensions. The pipeline and experiment artifacts that produced these findings are released with the report so subsequent work can build directly on top of them.

References

- [1] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.
- [2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [3] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. Connectionist

- temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *International Conference on Machine Learning (ICML)*, pages 369–376, 2006.
- [4] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *Association for Computational Linguistics (ACL)*, 2018.
 - [5] Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Uncertainty in Artificial Intelligence (UAI)*, 2018.
 - [6] Minghao Li, Tengchao Lv, Jingye Chen, Lei Cui, Yijuan Lu, Dinei Florencio, Cha Zhang, Zhoujun Li, and Furu Wei. TrOCR: Transformer-based optical character recognition with pre-trained models. In *AAAI Conference on Artificial Intelligence*, 2023.
 - [7] Yuting Li, Dexiong Chen, Tinglong Tang, and Xi Shen. HTR-VT: Handwritten text recognition with vision transformer. *Pattern Recognition*, 2024.
 - [8] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
 - [9] Urs-Viktor Marti and Horst Bunke. The IAM-database: an English sentence database for offline handwriting recognition. *International Journal on Document Analysis and Recognition (IJDAR)*, 5:39–46, 2002.
 - [10] Jamshed Memon, Maira Sami, Rizwan Ahmed Khan, and Mueen Uddin. Handwritten optical character recognition (OCR): A comprehensive systematic literature review (SLR). *IEEE Access*, 8:142642–142668, 2020.
 - [11] Konstantina Nikolaidou, George Retsinas, Vincent Christlein, Mathias Seuret, Giorgos Sfikas, Elisa Barney Smith, Hamam Mokayed, and Marcus Liwicki. WordStylist: Styled verbatim handwritten text generation with latent diffusion models. In *International Conference on Document Analysis and Recognition (ICDAR)*, 2023.
 - [12] Konstantina Nikolaidou, George Retsinas, Giorgos Sfikas, and Marcus Liwicki. Beyond memorization: Training-free style mixing for variability in handwritten text generation using writer embedding injection in pretrained diffusion models. In *International Conference on Document Analysis and Recognition (ICDAR)*, 2024.
 - [13] Georgios Retsinas, Giorgos Sfikas, Basilis Gatos, and Christophoros Nikou. Best practices for a handwritten text recognition system. In *International Workshop on Document Analysis Systems (DAS)*, pages 247–259, 2022.
 - [14] Mohamed Ali Souibgui, Alicia Fornés, Yousri Kessentini, and Beáta Megyesi. Interpretable writer recognition via vectors of locally aggregated characters. *Pattern Recognition Letters*, 2022.
 - [15] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.